Time blocking, Newton polygons and explicit quantum action

First calculation, 2026-10-27; written derivation, unrefereed. In one spatial coordinate and one time variable, the same discrete Green function controls three operations: constructing a Newton polygon on an arbitrary time partition, solving its nonlinear boundary-value problem, and bounding the inverse action Hessian used to integrate fluctuations. A force Lipschitz constant gives bounds uniform in the number and placement of inserted times. The quadratic fluctuation covariance contains an explicit factor of $\hbar$; this first calculation does not derive its positivity.

Targeted second opinion (GPT-6.1 Terra, 2026-10-27). Terra read this note and checked the Green row sum, Hessian/response bounds, Gaussian covariance, oscillator determinant and constant-force signs without finding an algebraic error. This was a targeted written check, not a full independent proof review. The arbitrary-cut interpretation below came from that pass.

The user directs Newton action necessity first, beginning with Balaban's techniques in $0+1$ dimensions. The comparison with Alpoge's work and Yang–Mills is deferred. Here $0+1$ means a mechanical degree of freedom along time, not a field with spatial lattice sites. The construction is a modern force polygon; no identification with every historical Newton update is asserted. The Galileo observable remains the inertial-line / falling-parabola defect, not swept sectors.

We assemble standard Green-function, contraction and Gaussian-elimination arguments. No literature novelty is claimed. The source ingredient is Balaban's controlled effective density under blocking, described in his ICM overview, printed pp. 1260–1262 (text read), and Dimock's small-field exposition, sections 2.1, 2.4, 4.5 and Theorem 14 (selected passages read). The existing refinement note constructs constant-force and harmonic limits. The cut-measure note, Proposition 7, already uses nodal exactness of the same one-dimensional Green function. The Rivero test, section 2, supplies the determinant/Schur identity and its branch caveat.

1. Actions and the constant to be investigated

Let $m, T > 0$, $x, y \in \mathbb{R}$, and $\pi = (0 = t_0 < \cdots < t_N = T)$. Write $\tau_j = t_{j+1} - t_j$ and $w_j = (\tau_{j-1} + \tau_j)/2$ for interior vertices. For a potential V define

$$S_\pi^\sigma(q) = \sum_{j=0}^{N-1} \frac{m(q_{j+1} - q_j)^2}{2\tau_j} + \sigma \sum_{j=0}^{N-1} \frac{\tau_j}{2}[V(q_j) + V(q_{j+1})], \qquad q_0 = x,\ q_N = y. \tag{1}$$

The real-time Newton action is $S_\pi^R = S_\pi^{-1}$, with weight $e^{iS_\pi^R/\hbar}$ when $\hbar > 0$. The Euclidean action is $S_\pi^E = S_\pi^{+1}$, with weight $e^{-S_\pi^E/\hbar}$. Their stationary equations have opposite potential signs. They must not be interchanged when interpreting a background as a Newton trajectory. At $\hbar = 0$ these exponential expressions are not defined; the classical stationary equation is defined directly. No relativistic premise or c is used in this calculation. Ordinary Planck action is $h_P = 2\pi\hbar$.

For real time put $f = -V'/m$, the acceleration. Section 2 needs only a Lipschitz function f, even without first specifying V. A Lipschitz condition is a sufficient uniqueness and stability condition; continuity alone can give local existence. Neither statement alone guarantees continuation for all times or uniqueness of a two-endpoint problem.

2. Arbitrary intermediate times

Let $\ell(t) = x + (y - x)t/T$ and

$$G(t, s) = \min(t, s) - \frac{ts}{T}. \tag{2}$$

This is the positive Dirichlet Green function of $-d^2/dt^2$. Define the polygon map, with piecewise-linear output, by

$$(\mathcal{T}_\pi q)(t) = \ell(t) - \sum_{j=1}^{N-1} w_j G(t, t_j) f(q(t_j)). \tag{3}$$

Lemma 1 (the exact row sum). At every vertex,

$$\sum_{j=1}^{N-1} w_j G(t_i, t_j) = \frac{t_i(T - t_i)}{2}. \tag{4}$$

At every $t \in [0, T]$ the sum is at most $T^2/8$.

Proof. For fixed t_i, the function $s \mapsto G(t_i, s)$ is linear on each partition interval and vanishes at the endpoints. Its trapezoidal integral is therefore exact, and its integral is $t_i(T - t_i)/2$. As a function of t, the sum in (4) is the linear interpolant of that concave quadratic. It lies below the quadratic, whose maximum is $T^2/8$. This proof uses no ratio between consecutive time steps. □

Assume $|f(u) - f(v)| \le \Lambda|u - v|$ for all u, v, and put

$$\theta = \frac{\Lambda T^2}{8} < 1. \tag{5}$$

Proposition 2 (one bound for every polygon). The map (3) has a unique fixed polygon. Its slopes $v_j = (q_{j+1} - q_j)/\tau_j$ satisfy

$$v_j - v_{j-1} = w_j f(q_j). \tag{6}$$

The continuous map

$$(\mathcal{T} q)(t) = \ell(t) - \int_0^T G(t, s) f(q(s))\, ds \tag{7}$$

has a unique fixed point, which is C^2 and solves $\ddot{q} = f(q)$ with the two prescribed endpoints. Every sequence of partitions with $|\pi| \to 0$ converges to this solution. Neither nesting nor equal intervals is required.

Proof. Lemma 1 bounds the Lipschitz constants of (3) and (7) in the supremum norm by θ. Repeated application converges because the successive differences form a geometric series. This proves existence and uniqueness in the complete polygon space or in $C([0, T])$. The derivative of $G(\cdot, t_j)$ jumps by -1 at t_j, giving (6). Differentiating (7) gives the continuous equation.

Inserting a vertex changes its neighbours' weights w_j as well as the new solution. It does not add one kick while freezing the old force history. Retained vertices need not remain fixed when the trapezoidal action itself is refined.

For a quantitative limit, write $B = \|f(q)\|_\infty$ and $U = \|\dot{q}\|_\infty$ for the continuous solution. For every fixed t, $s \mapsto G(t, s) f(q(s))$ is Lipschitz with constant at most $B + T\Lambda U/4$. On an interval of length τ, replacing its integral by the endpoint trapezoid has error at most half that constant times τ^2. Hence

$$\|q_\pi - q\|_\infty \le \frac{T|\pi|}{2(1 - \theta)} \left(B + \frac{T\Lambda U}{4} \right). \tag{8}$$

Indeed, subtract the two fixed-point equations and absorb the $\theta\|q_\pi - q\|_\infty$ term. The constants in (8) belong to the fixed solution and interval, not to the partition. □

For a force Lipschitz only in a tube $|q(t) - \ell(t)| \le r$, the same proof holds within that tube if $\theta < 1$ and

$$\frac{T^2}{8}(B_0 + \Lambda r) \le r, \qquad B_0 = \sup_t |f(\ell(t))|. \tag{9}$$

This makes the closed tube invariant. Uniqueness is then asserted only inside it. Equation (9) is a finite version of a small-field domain that survives the construction; it cannot justify discarding paths outside it from a statistical or oscillatory integral.

3. The action Hessian and elimination

Let K_π be the Dirichlet stiffness matrix, with diagonal $1/\tau_{j-1}+1/\tau_j$ and adjacent entries $-1/\tau_j$, and let $\mathsf{W}_\pi = \operatorname{diag}(w_j)$. The tent function (2) gives

$$(\mathsf{K}_\pi^{-1})_{ij} = G(t_i, t_j), \qquad \|\mathsf{K}_\pi^{-1}\mathsf{W}_\pi\|_{\infty\to\infty} \le \frac{T^2}{8}. \tag{10}$$

The first identity follows by taking the slope jump of that tent at each node; the second is Lemma 1. In particular, for $V \in C^2$ with $|V''| \le m\Lambda$, the Hessian of (1) obeys

$$\mathsf{H}_\pi^\sigma = m\mathsf{K}_\pi + \sigma\mathsf{W}_\pi \operatorname{diag}(V''(q_j)) \;\ge\; m(1-\theta)\mathsf{K}_\pi. \tag{11}$$

To see the quadratic-form inequality, the nonnegative matrix $\mathsf{K}_\pi^{-1}\mathsf{W}_\pi$ has spectral radius at most its maximum row sum. It is similar to the positive symmetric matrix $\mathsf{W}_\pi^{1/2}\mathsf{K}_\pi^{-1}\mathsf{W}_\pi^{1/2}$. Thus $\mathsf{W}_\pi \le (T^2/8)\mathsf{K}_\pi$, which proves (11). The factor $1-\theta$ is uniform over arbitrary partitions.

For a physical force source J_j, inserted as a linear action term $\sum_j w_j J_j q_j$, the stationary response therefore has the bound

$$\|\delta q\|_\infty \le \frac{T^2}{8m(1-\theta)}\|\delta J\|_\infty. \tag{12}$$

Here the bound is for the differential response. It follows directly by factoring H_π^σ as $m\mathsf{K}_\pi(I + \mathsf{K}_\pi^{-1}\mathsf{W}_\pi\sigma \operatorname{diag}(V'')/m)$ and using a Neumann series.

With the global curvature bound, (11) makes the finite action coercive and strictly convex in its interior vertices. Eliminating some vertices by minimization is then unambiguous, and its Hessian is the Schur complement

$$\mathsf{H}_{BB} - \mathsf{H}_{BA}\mathsf{H}_{AA}^{-1}\mathsf{H}_{AB}. \tag{13}$$

Successive eliminations associate. With only a local tube, this is a statement on the chosen nondegenerate branch. In particular, for the real-time positive quartic potential the unrestricted action is unbounded below, as the Rivero note already records; local control is not a global minimization theorem. Nor does the eliminated action generally retain only the original kinetic and potential terms.

4. What the same inverse says when $\hbar$ is present

At a Euclidean background, its Gaussian fluctuation reference has density proportional to $\exp[-\eta^T\mathsf{H}_\pi^E\eta/(2\hbar)]$ and covariance

$$\mathsf{C}_{\pi,\hbar}^E = \hbar(\mathsf{H}_\pi^E)^{-1}, \qquad (\mathsf{C}_{\pi,\hbar}^E)_{jj} \le \frac{\hbar}{m(1-\theta)}\frac{t_j(T-t_j)}{T} \le \frac{\hbar T}{4m(1-\theta)}. \tag{14}$$

These are exact Gaussian statements. For a nonlinear action they do not assert the covariance bound for its full measure without a further argument. In real time, normalized quadratic Fresnel integration gives the algebraic centered second moment $i\hbar(\mathsf{H}_\pi^R)^{-1}$, which is not a probability covariance.

For one block of Gaussian variables, integration retains both the Schur complement (13) and a determinant factor: respectively $(2\pi\hbar)^{|A|/2}(\det \mathsf{H}^E_{AA})^{-1/2}$ in Euclidean time and $(2\pi i\hbar)^{|A|/2}(\det \mathsf{H}^R_{AA})^{-1/2}$ when the real Hessian is positive. Determinants, normalization and any higher interactions must accompany the stationary action. These are the extra data that Balaban's effective-density viewpoint instructs us to retain.

Equations (11)–(14) identify the connection to ODE control: the curvature constant that makes the background construction stable also controls the Gaussian fluctuations for every positive supplied $\hbar$. They do not create the multiplier $\hbar$ in (14).

5. Written controls: Galileo and the oscillator

Constant force. For $V(q) = -Fq$, $\Lambda = 0$. Equation (7) is exactly

$$q(t) = \ell(t) + \frac{F}{2m}t(t-T). \tag{15}$$

By (4), every polygon has these same vertex positions. The fluctuation Hessian is the free one for either action sign, and the Euclidean covariance is $(\hbar/m)G(t_i,t_j)$. For a single inserted midpoint of a cell of length T, its variance is $\hbar T/(4m)$; for an unequal cut at u it is $\hbar u(T-u)/(mT)$. The deterministic vertex survives at $\hbar = 0$; the Gaussian width tends to zero. This uses precisely the Galileo comparison already normalized in the refinement note.

One arbitrary cut (Terra's connection). Split $T = u + v$ and let H_u be the piecewise-linear hat of height one at u, zero at the cell endpoints. Then

$$\int_0^T (H_u')^2 dt = \frac{T}{uv}, \qquad a_F = -\frac{Fuv}{2m}, \qquad \frac{m}{2}\int_0^T (a_F H_u')^2 dt = \frac{F^2 uvT}{8m} = K_T - K_u - K_v. \tag{15a}$$

where $K_s = F^2 s^3/(24m)$. The free bridge's conditional inserted-vertex innovation has variance $s_u^2 = \hbar uv/(mT)$. For the two equal-variance Gaussian vertex laws, with means separated by a_F, direct integration of their log-density ratio gives

$$D\big(N(a_F, s_u^2)\,\|\,N(0,s_u^2)\big) = \frac{a_F^2}{2s_u^2} = \frac{K_T - K_u - K_v}{\hbar}. \tag{15b}$$

Here D is relative entropy in nats. Successive bridge innovations factor into independent Gaussian hat coefficients: conditioned on a cell's endpoints, its inserted value has the variance above, and the two residual bridges are independent. The hats are also orthogonal in the Dirichlet energy because each finer hat has zero derivative integral inside a cell where an older hat has constant derivative.

Thus classical vertex elimination, conditional Gaussian normalization and the cut energy are related by one Schur factorization. The cut-measure note already proves the additive action budget; the useful connection here is to the blocking variables. Equation (15b) assumes these Gaussian laws. It does not establish a physical bound on the information in complete apparatus records.

Oscillator and caustics. Let $V = m\omega^2 q^2/2$, $\omega > 0$. Its initial-value equation has the unique explicit solution $q(t) = q(0)\cos\omega t + \dot q(0)\sin\omega t/\omega$ for all times. Nevertheless its two-endpoint problem loses uniqueness at $T = \pi/\omega$ when $y = -x$; for other y it has no solution at that time. A singular boundary Hessian therefore need not mean a failure of initial-value mechanics.

On a uniform N-cell partition, $\tau = T/N$, $N \geq 2$, the first Dirichlet eigenvalue of the difference operator is $4\tau^{-2}\sin^2(\pi/(2N))$. Its Hessian first becomes singular at

$$T_N = \frac{2N}{\omega}\sin\frac{\pi}{2N} \longrightarrow \frac{\pi}{\omega}. \tag{16}$$

This checks the partition-independent sufficient window $\omega^2 T^2 < 8$: $T_2 = \sqrt{8}/\omega$, while the continuum caustic is later. An ODE Lipschitz constant cannot exclude those later caustics.

For an exact Gaussian control, suppose $T < \sqrt{8}/\omega$ and set $\alpha = 2\arcsin(\omega\tau/2)$. The determinant ratio is

$$R_N = \frac{\det \mathsf{H}_\pi^R}{\det(m\mathsf{K}_\pi)} = \frac{\sin(N\alpha)}{N\sin\alpha} \longrightarrow \frac{\sin(\omega T)}{\omega T}. \tag{17}$$

Indeed the determinant of the tridiagonal matrix with diagonal $2 - \omega^2\tau^2 = 2\cos\alpha$ and off-diagonal -1 satisfies $D_n = 2\cos\alpha D_{n-1} - D_{n-2}$, with $D_0 = 1, D_1 = 2\cos\alpha$. Thus $D_{N-1} = \sin(N\alpha)/\sin\alpha$; the free determinant is N. For $N = 2$, (17) gives $R_2 = 1 - \omega^2\tau^2/2$, also obtained by completing the single midpoint square.

With the usual free normalization for each cell, finite Gaussian integration gives the real-time kernel

$$K_\hbar^\pi(y, x) = \sqrt{\frac{m}{2\pi i\hbar T}}\, R_N^{-1/2} e^{iS_\pi^R(q_\pi)/\hbar}. \tag{18}$$

The prefactor follows from $\det \mathsf{K}_\pi = T/\prod_j \tau_j$ and completing the quadratic form. The determinant identity follows by the same tridiagonal recurrence, or by successive free one-vertex eliminations. Equation (18) retains the same $\hbar$ at every depth. Reversing the sign of the oscillator term would put $1 + \omega^2\tau^2/2$ in the two-cell determinant and remove the real-time conjugate point. This sign test distinguishes the Euclidean and Newton constructions.

6. Large fields, continuation and the next scale question

A local Lipschitz bound controls a local construction. To continue an initial-value solution, one also needs control preventing escape from every bounded phase-space region. In the confining example $V(q) = \lambda q^4/4$, $\lambda > 0$, conservation of a fixed classical energy $E > 0$ gives

$$|q| \le R_E = (4E/\lambda)^{1/4}, \qquad |p| \le \sqrt{2mE}, \qquad |f'| \le \frac{3\lambda R_E^2}{m}. \tag{19}$$

These bounds permit continuation of the initial-value solution. Applying the local polygon lemma to a whole tube may require a larger radius than R_E and the invariance condition (9). For the inverted quartic, $q(t) = q_0/[1 - q_0\sqrt{\lambda/(2m)}t]$, $q_0 > 0$, is instead a finite-time escape solution. Its force is locally Lipschitz everywhere. This is an explicit reason to distinguish local control from continuation. Balaban's large-field estimates address weighted integrals over all field regions; that is a statistical obligation distinct from trajectory continuation.

Keeping $\hbar$ explicit in the positive quartic Euclidean action, set $t = Ts$ and $q = \sqrt{\hbar T/m}\, z$. Then

$$\frac{S^E}{\hbar} = \int_0^1 \left[\frac{(z')^2}{2} + \frac{g}{4}z^4\right] ds, \qquad g = \frac{\lambda\hbar T^3}{m^2}. \tag{20}$$

For the zero-endpoint free reference, the Brownian-bridge reflection bound is

$$\mathbb{P}\Big(\sup|q| > b\sqrt{\hbar T/m}\Big) \le 2e^{-2b^2}, \qquad b > 0. \tag{21}$$

It follows by applying the positive and negative one-sided reflection bounds and a union bound; the Rivero note, section 3, supplies the same calculation. On this small-field tube the real quartic curvature gives $\theta \le 3b^2 g/8$. Thus the natural perturbation parameter combines force curvature, the observation interval and the explicitly supplied quantum width. For the interacting normalized measure, its denominator and its own large-field contribution still require control; (21) alone is not that result.

The next useful test is to retain the normalized fluctuation and large-field terms through repeated time insertion, and determine which physical premise would fix their action multiplier independently of the weight $e^{-S^E/\hbar}$ or $e^{iS^R/\hbar}$. The existing quartic zero-limit theorem prevents treating refinement consistency alone as that premise. A loss of boundary invertibility also needs the oscillator countertest above before it is used to argue for completion of Newton's dynamics.

A complementary construction suggested by Terra is to control the initial-value polygon flow and its tangent map under arbitrary insertion. Their composition is causal and remains meaningful through the oscillator's boundary caustics. Comparing that description with the endpoint action would separate loss of one coordinate description from an actual need for additional physical dynamics.

7. Consequence for STATE

Newton action necessity is the active priority. This note gives the first 0+1 background construction: arbitrary unequal polygons converge on a stated Lipschitz-controlled interval; the same Green function controls source response and the Gaussian width proportional to $\hbar$. Its oscillator example separates boundary caustics from initial-value failure. These are written, unrefereed calculations. Next retain the full weighted corrections and justify their physical weighting without importing the positive action constant to be derived. Alpoge and Yang–Mills remain for a later comparison.